

Waypoint Group - Delivery Planning System: UI/UX Wireframe Spec

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Tech-Triathlon 2026 Designathon submission: multi-role delivery planning system wireframe specification covering Dispatcher, Loader, Driver, and Store Manager roles, plus the shared time-based theme system.

1. Global Design System

1.1 AM/PM Adaptive Theme

Theme switches automatically on device time; a manual override toggle sits in the header badge for dispatchers working across shift boundaries.

Morning Mode — 3:30 AM to 11:59 AM (light, high-contrast, glare-safe for docks and outdoor driving)

Token	Hex	Usage
--bg-base	#F7F9FB	App background
--surface	#FFFFFF	Cards, panels
--text-primary	#12181F	Body text (contrast ratio 15.8:1 on surface)
--text-secondary	#4B5563	Labels, meta text
--accent-urgent	#D63A2E	Fresh-delivery urgency (before 8 AM)
--accent-primary	#0F6FDE	Primary actions, links
--success	#1E8E5A	Confirmed / on-time states
--warning	#C77700	Deferral / delay states
--border	#DDE3EA	Dividers, input borders

Afternoon/Evening Mode — 12:00 PM to 11:59 PM (dark, warm-toned amber/navy, eye-strain reduction)

Token	Hex	Usage
--bg-base	#151A22	App background
--surface	#1E2530	Cards, panels
--text-primary	#F0E9DD	Body text (contrast ratio 12.6:1 on surface)
--text-secondary	#B7A98F	Labels, meta text
--accent-urgent	#E8A33D	Fresh-delivery urgency, evening ramp alerts
--accent-primary	#4C8DFF	Primary actions, links
--success	#4CAF7D	Confirmed / on-time states
--warning	#E8A33D	Deferral / delay states
--border	#2E3746	Dividers, input borders

Typography — Inter (UI text), JetBrains Mono (IDs, coordinates, timestamps)

Style	Size / Weight	Use
Display	28px / 700	Dashboard KPI numbers
H1	22px / 700	Screen titles
H2	17px / 600	Section headers
Body	15px / 400	Default text
Body Small	13px / 400	Meta, secondary labels
Caption	11px / 500, uppercase, tracked +0.04em	Status chips, tags

Minimum tap target 44×44px on tablet/mobile roles (Loader, Driver, Store Manager mobile view); minimum body text 15px, scalable to 18px in Driver safe-stop view.

Contextual Theme Header Badge (all roles): pill-shaped, top-right of nav bar. Shows sun/moon glyph + "Morning Mode" / "Evening Mode" text + a live countdown "Fresh cutoff in 1h 42m" or "Next window closes 6:15 PM". Tapping expands a mini-panel with the manual override switch.

1.2 Integrated Logistic Calendar & Events System

Shared calendar component (`<CalendarWidget>`) surfaces on every role's dashboard as a compact strip, expandable to full month view.

- **Data sources merged:** competition calendar feed (paydays, monsoon windows, festivals, public holidays) + live weather alerts + internal ops calendar (blackout dates, fleet maintenance).
- **Compact strip (default):** horizontal 7-day scroller, each day a chip colored by event severity (grey = normal, amber = advisory, red = high-impact). Today is highlighted with a ring.
- **Full month view:** standard month grid; tapping a marked day opens a bottom-sheet (mobile) or side panel (desktop) listing that day's events with icons (🎉 festival, 🌧️ monsoon, 💰 payday, 🏛️ holiday).
- **Role-based contextual alert card:** appears above the fold on the dashboard when an event is within 72 hours. Format: `[Icon] [Event Name] – [Impact statement] · [Dismiss] [View Plan]`.
 - Example (Dispatcher): "🎉 Vesak Festival Ramp Active — Expect +40% Fresh Demand across Colombo District. [View Fleet Plan]"
 - Example (Driver): "🌧️ Monsoon Heavy Rain Alert — Kandy District. Reduced speed advised on Route A9. [Acknowledge]"
 - Example (Store Manager): "💰 Payday Window (25th–27th) — Expect higher order volume. Confirm cutoff early. [Noted]"
- Alerts are dismissible per-user but reappear each session until the event passes; high-impact (red) alerts cannot be permanently dismissed, only acknowledged.

2. Role 1 — Dispatcher (Peliyagoda Office, large-screen desktop)

2.1 Layout: Central Planning Dashboard

Three-column desktop grid (1920×1080 baseline): **Left rail (240px)** nav + theme badge; **Center (flex)** Consolidated Queue + Fleet Allocation Engine tabs; **Right rail (360px)** Live Network Map (collapsible) + Calendar strip.

- Top bar: Logo · Search orders/outlets · Theme badge ("☀ Morning Mode · Fresh cutoff in 0h 38m") · Notification bell · Profile.
- Tab strip below top bar: `Queue` | `Fleet Allocation` | `Deferrals` | `Live Map` | `Calendar` .

2.2 Consolidated Queue (post-4PM cutoff)

Table view, one row per confirmed order: Order ID · Outlet · District · Weight (kg) · Volume (m³) · Temp Req (Reefer/Ambient chip) · Priority (Fresh/Standard) · Status (Unassigned/Assigned/Deferred).

- Header actions: [Auto-Assign All] · [Filter by District ▼] · [Filter by Temp ▼] · sort by priority (default).
- Row click opens a slide-over with full order detail + manual assign control.
- Empty state copy: "No confirmed orders yet — queue populates after the 4:00 PM cutoff."

2.3 Fleet Allocation Engine

Split view: left = order queue (draggable cards), right = vehicle roster cards (Van 04 — 800kg / 3.2m³ — Reefer — Driver: K. Perera — Shift ends 8:00 PM — Fuel: 62%).

- **Auto** mode: button [Run Auto-Assignment] — solves against capacity, temp compatibility, van-only access flags, fuel quota, and remaining shift budget; result shown as a diff ("14 orders assigned, 2 flagged — see below").
- **Manual** mode: drag order card onto a vehicle card; system blocks the drop and shows an inline red banner if a constraint is violated: "⚠ Exceeds Van 04 capacity by 45kg" / "⚠ Reefer required — Van 04 is Ambient-only" / "⚠ Driver shift ends in 20 min — insufficient time for this stop."
- Toggle switch top-right: **Auto** ⇌ **Manual**.
- Confirm bar (sticky bottom): [Lock Fleet Plan] — disabled until all high-priority orders are assigned or explicitly deferred.

2.4 Deferral Management Panel

Modal triggered from an order's slide-over via [Defer Order].

- Fields: Reason (dropdown: Capacity exceeded / Access restriction / Weather disruption / Driver shift limit / Other) · Notes (free text, required if "Other") · Reschedule to (date picker, defaults to next business day).
- Confirm button: [Confirm Deferral]; cancel: [Back to Queue].
- Below the modal trigger, a persistent panel lists **Deferral History**: Outlet · Consecutive Skips (badge turns red at 3+) · Last Reason · [View Trend].

- Alert copy when an outlet crosses 3 consecutive skips: "⚠ Colombo Fort Pharmacy has been deferred 3 times running — escalate to Account Manager?"

2.5 Live Network Map

Right-rail embedded map, expandable to full-screen.

- Layers toggle: Drivers · Traffic · Disruptions · Route Lines .
- Driver pins are colored by status (green = on schedule, amber = delayed, red = stopped >10 min unexpectedly) and show a live speed index badge.
- Disruption pins (road closure, flooding) pulled from the calendar/weather feed; tapping shows a card: "🚧 Kandy-Peradeniya Rd — partial closure, +25 min detour advised."
- Bottom strip: horizontally scrollable driver chips (K. Perera · Van 04 · ETA next stop 14:20) — tapping centers the map on that driver.

2.6 Calendar & Demand Forecast Integration

Uses the shared `<CalendarWidget>` (§1.2), plus a **Demand Forecast overlay**: each calendar day shows a small bar indicating projected order volume vs. average. Festival/monsoon days show the projected % uplift or capacity reduction inline (" +40%", "-15% capacity"). A **[Adjust Fleet for This Day]** shortcut jumps to Fleet Allocation pre-filtered to that date.

3. Role 2 — Loader (Peliyagoda/Kandy Docks, shared tablet)

High-contrast, gloves-usable UI: large tap targets (min 56×56px), bold 2px iconography, dock-mode always-on Morning palette (regardless of clock) since dock lighting is artificial and glare-prone.

3.1 Layout: Digital Loading Manifest (home screen)

Single-column, full-width cards, one per truck run.

- Card header: Run #204 — Van 04 — Dock Bay 3 + live sync indicator (green dot "Synced" / amber "Updating...").
- Card body: progress bar "7 of 12 items loaded" + **[Open Stop Sequence]** primary button (full-width, 64px tall).

- Manifest auto-refreshes when the dispatcher updates the run plan; a toast slides in: "Run plan updated — 2 items added, tap to review."

3.2 Stop Sequence Visualizer

Vertical numbered list mirroring unload order (last-loaded = first-unloaded logic reversed for load order: items for the FINAL stop load first, at the back of the van).

- Each row: [Stop #] [Outlet Name] [Item Count] [Load Position: Front/Mid/Rear] with a checkbox.
- Color coding: grey = pending, blue = in progress, green = loaded & verified.
- Sticky bottom bar: **[Mark Stop Loaded]** — disabled until every item checkbox in that stop is ticked or explicitly flagged (§3.3).

3.3 Pre-Departure Shortfall & Damage Logger

Triggered by long-press on an item row, or the **[Flag Item]** icon.

- Bottom-sheet form: Issue Type (large icon buttons:  Missing /  Damaged /  Quantity Mismatch) → Photo (camera capture, required for Damaged) → Notes (optional voice-to-text supported) → **[Submit Flag]**.
- Flagged items show a red badge on the manifest and stay in the count as "Exception" rather than blocking the whole run.
- Confirmation toast: "Flag logged — Dispatcher notified."

3.4 Pre-Departure Sign-off

Full-screen final screen before a run can depart: summary "11 of 12 items loaded · 1 flagged (Missing)" + **[Confirm Load Complete]** (large, primary, bottom) + secondary **[Review Exceptions]**. Requires loader's PIN or badge tap to confirm — logs Loader ID, Timestamp, Dock.

4. Role 3 — Driver (On-road, smartphone, vertical layout)

Safety-first: no interaction requires more than one glance while stationary; "drive mode" locks most controls when GPS speed > 8 km/h, surfacing only voice/audio controls.

4.1 Safe-Stop Navigation View (default screen)

Full-screen map (70% viewport), bottom sheet (30%, expandable) with next-action card.

- Map: current position pin, route polyline, upcoming stop pins numbered in sequence, depot pin at route start.
- Bottom sheet collapsed state: [Stop 4 of 9] Kandy City Pharmacy – 2.3 km · ETA 14:32 + big **[Navigate]** button.
- When device is moving, sheet auto-collapses to a single-line strip; when parked (speed = 0 for >3 sec), sheet expands and unlocks full controls — a subtle banner reads "Parked — full controls unlocked."

4.2 Smart Call & Notification Reminders

- Incoming alerts render as a top banner overlay (never blocking the map), auto-read aloud via TTS when moving: "Dispatcher calling" / "Outlet Manager arriving in 5 minutes" / "Stop window closes in 15 minutes."
- Hands-free actions: swipe-right-to-accept-call (haptic confirm), voice command "Hey Waypoint, call dispatcher" wakes a call sheet.
- Stop-window warning escalates: 15 min = amber banner + single chime; 5 min = red banner + repeating chime + auto-suggested "Call outlet" action.

4.3 Arrival & Delivery Execution

- On approach (geofence trigger, ~150m), sheet expands: **[I've Arrived]** primary button.
- Post-arrival: "Delivery Checklist" card — access constraints surfaced up top in a highlighted strip: "⚠ Rear dock access only" / "⚠ Mall access: 9 AM–9 PM, use Loading Bay B."
- Next-destination preview strip pinned below: "Next: Stop 5 — Peradeniya Mini Mart — 4.1 km" so the driver can plan before finishing current stop.

4.4 Proof of Delivery (POD) & Issue Capture

- Screen order: 1) Item checklist (tap to confirm each line) → 2) **Signature pad** (finger-draw, "Recipient Name" text field above it) → 3) **Photo capture** (auto, min 1 required, camera opens directly) → 4) Discrepancy toggle: "Everything OK" / "Report Issue" (opens same Issue Type picker as Loader's flag flow: Missing / Damaged / Refused) → **[Complete Delivery]**.
- Confirmation screen: green check animation + "Stop 4 complete — 5 remaining" + auto-advances to next-stop nav after 3 sec.

4.5 Offline Mode & Local Storage

- Persistent status chip in top bar: "● Online" / "● Offline — 3 actions queued" / "● No signal since 14:10."
- All POD, arrival, and issue actions write to local storage first and queue for sync; nothing blocks on network.
- On reconnect: toast "Reconnected — syncing 3 updates..." followed by "✓ All updates synced" or, on conflict (e.g., dispatcher re-routed while offline), a review sheet: "Route changed while you were offline — 1 stop added, 1 removed. [Review Changes]".
- Full offline map caching pre-downloaded for the day's route corridor (esp. Hill Country/Kandy corridor) at shift start, shown via a "Offline maps ready for today's route ✓" confirmation on login.

5. Role 4 — Store Manager (Outlets, desktop/mobile responsive)

Responsive single-codebase layout: desktop = 2-column (main + side panel), mobile = stacked single column with bottom tab bar (**Orders** · **Track** · **Deferred** · **Receipts** · **Calendar**).

5.1 Order Entry & Confirmation Desk

- Form: **Product** (searchable catalog picker) → **Quantity** → **Add to Cart**; running cart summary panel on the right (desktop) / collapsible drawer (mobile).
- Cutoff indicator pinned above the submit button: "Cutoff 4:00 PM — 1h 12m remaining" (turns red under 15 min).
- Submit button: **[Place Order]** → immediate confirmation card: "✓ Order #4471 confirmed — Expected delivery window 9:00–11:00 AM tomorrow."
- Past-cutoff state: submit button becomes **[Add to Next Cycle]** with copy "Cutoff passed — this order joins tomorrow's 4 PM cycle."

5.2 Live Delivery Status & ETA Tracker

- Card per active order: **Order #4471** · status pill (Assigned / En Route / Arriving / Delivered) · mini-map thumbnail with live driver pin · "Driver: K. Perera · ETA 10:42 AM (updates live)".
- Tapping expands to full map + stop sequence context: "You are stop 4 of 9 on this run."
- Push/in-app notification at 3 stops out: "Your delivery is 3 stops away — estimated arrival 10:40 AM."

5.3 Deferred Order Notification Hub

- Dedicated tab, list view: **Order ID** · **Original ETA** · **New ETA** · **Reason** (plain-language, e.g. "Vehicle capacity — reassigned to next available run") · **[View Details]**.
- Top-of-list banner on new deferral: "⚠ Order #4471 has been rescheduled — new window 2:00–4:00 PM today."
- Each entry has **[Acknowledge]** to clear the unread badge, and **[Contact Dispatch]** for disputes.

5.4 Receipt Confirmation & Claim Portal

- On driver-marked delivery, an in-app prompt fires: "Delivery arrived — confirm receipt" → checklist mirroring the driver's manifest → **[Confirm Receipt]** (digital signature capture) or **[Report Discrepancy]**.
- Discrepancy flow: same Issue Type picker (Missing/Damaged/Wrong Item) → photo upload → notes → **[Submit Claim]** → generates **Claim #C-4471-01** with tracked status (Submitted → Under Review → Resolved).
- Digital receipt auto-generated as a PDF-style card: itemized list, driver name, timestamp, signature thumbnail — **[Download Receipt]**.

5.5 Special Calendar View

Extends the shared `<CalendarWidget>` (§1.2) with an outlet-specific promotional layer: days highlighted for store-specific festival promotions or bulk-delivery scheduling, each with a **[Pre-order for this event]** shortcut that jumps to Order Entry with a suggested quantity multiplier pre-filled (e.g. "+40% suggested for Vesak week").

6. Degradation & Failure Scenario Wireframes

6.1 Driver Offline Recovery Flow

flowchart TD

```

A[Signal lost] --> B[Status chip: Offline]
B --> C[Driver continues: arrivals, POD, issues all queue locally]
C --> D{Signal restored?}
D -- No --> C
D -- Yes --> E[Toast: Reconnected - syncing N updates]
E --> F{Conflicts with dispatcher changes?}
F -- No --> G[Toast: All updates synced]
F -- Yes --> H[Review Changes sheet shown]
H --> I[Driver taps Acknowledge]
I --> G

```

Screen specs:

- *Offline banner* (persistent, non-blocking, top of screen): amber background, "🟡 Offline since 14:10 — 3 actions saved locally, will sync automatically."
- *Sync-in-progress overlay* (brief, ~2 sec): spinner + "Syncing 3 updates..." — does not block map interaction underneath.
- *Conflict review sheet* (only when the dispatcher altered the route during the offline window): title "Your route changed while offline", body list + Added: Stop 6 — Gampola Grocers / - Removed: Stop 3 — deferred by dispatcher, single button **[Got it — Update My Route]**.
- *Data integrity rule*: every locally-queued action carries a client-generated UUID + timestamp so duplicate syncs are rejected server-side (idempotent write) — no wireframe surface for this, but it governs what "synced" is allowed to mean.

6.2 Loading Shortfall Flow (Loader)

flowchart TD

```

    A[Loader ticks item] --> B{Item present & undamaged?}
    B -- Yes --> C[Green check, manifest count updates]
    B -- No --> D[Long-press / Flag Item]
    D --> E[Issue Type sheet: Missing / Damaged / Quantity Mismatch]
    E --> F[Photo required if Damaged]
    F --> G[Submit Flag]
    G --> H[Item marked Exception – red badge]
    H --> I[Dispatcher notified in real time]
    I --> J{Run can still depart?}
    J -- Yes, exceptions noted --> K[Confirm Load Complete]
    J -- No, critical shortfall --> L[Dispatcher re-run Fleet Allocation for affected stop]
  
```

Screen specs:

- *Exception badge*: red dot + count in the manifest card header, e.g. "Run #204 — 1 exception."
- *Dispatcher-side mirror*: the Fleet Allocation Engine (§2.3) shows a live toast the moment a flag is submitted: "⚠ Van 04 — Item flagged Missing (SKU 88213) — outlet: Kandy Pharmacy." Dispatcher can either accept the shortfall (outlet notified automatically via the Deferred Order hub, §5.3, if the missing item is order-critical) or trigger a re-allocation.
- *Store Manager-side mirror*: if a flagged shortfall affects their order, it surfaces as a Deferred Order Notification (§5.3) with reason auto-filled: "Partial shortfall at dispatch — 1 item unavailable, rest of order proceeding as scheduled."
- *Sign-off gate*: **[Confirm Load Complete]** remains enabled with exceptions present (a full stop is not required for a single flagged item), but is disabled entirely if a *critical* item (flagged "Quantity Mismatch" on a Fresh-priority order) is unresolved — forcing dispatcher intervention before departure.